TALK PLAN / PRINT THIS PAGE AS YOUR RUN SHEET

Cutoff for exclusion with inhomogeneous conductances

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Format: 43 minutes of presentation + 2 minutes for questions. English; aimed at probability graduate students and faculty. Eight successive board panels, with spoken explanations accompanying the formulas.

Contribution to remember. Under the stated resistance and particle-number assumptions, spatially varying rates preserve the homogeneous cutoff time, including its leading constant. The explanation is a slow mode that starts at size k and reaches equilibrium fluctuations of size $\sqrt{k}$:

$$k \exp(-\lambda_1 t) \asymp \sqrt{k} \quad \Longrightarrow \quad t \sim \frac{N^2}{2\pi^2} \log k.$$

Clock	Board / topic	Audience should take away
00–05	1. The dynamics	Disorder changes rates; equilibrium stays uniform.
05–11	2. Cutoff and the theorem	The full theorem and the meaning of its assumptions.
11–15	3. What is new?	Sharp cutoff beyond homogeneous rates; the new work.
15–20	4. The spectral input	What the gap, shape, and flux each contribute.
20–27	5. Why a cosine?	The Riccati recurrence, its pole, and the Neumann condition.
27–34	6. Lower bound	A detectable slow mode explains the factor 1/2.
34–40	7. Upper-bound geometry	Weights adapted to the environment.
40–43	8. Conclusion	Why coupling completes the theorem; contributions.
43–45	Questions	Stop writing; leave the time scale visible.

How to use these notes

Pages 2–9 are the talk. Shaded boxes contain the formulas to copy; the prose supplies the spoken explanation. The backup pages contain longer calculations, attribution, and answers to questions. Rehearse the transitions, not every sentence verbatim.

Scope. The slides are the main source; the spectral paper supplies Proposition 1.1 and Theorem 1.3. The talk retains the full cutoff hypotheses and explains the lower bound in that setting. The technical coalescence estimate is stated as an input. The spectral theorem uses resistance averaging alone.

Board management

Keep a narrow left-hand strip for $r_x = 1/c_x$, $\mu_{N,k}$, $\lambda_1 \sim \pi^2/N^2$, and the boxed cutoff time. Use the remaining space for the current panel. Copying a shaded box should take at most two minutes; explain its meaning as you write.

If you are running late: explain the reciprocal change of variable on Board 5 verbally and give the drift on Board 7 verbally. Preserve Board 3: the audience should understand the contribution before hearing the proof.

BOARD 1 / 00–05 MINUTES

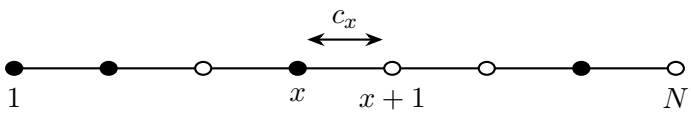
A disordered dynamics with a simple equilibrium

Say

Imagine that all the particles start on the left of a segment. Eventually they spread out and forget that initial arrangement. For identical jump rates, the time of this transition is known precisely. Today I will explain what happens when each edge has its own rate. Under suitable averaging and resistance bounds, the same sharp transition survives, with the same leading constant.

Draw first

Draw a segment with occupied and vacant sites, label one edge c_x, and draw a two-way arrow above it. Explain that there are no edges beyond the endpoints.



WRITE ON THE BOARD

$$\Omega_{N,k} = \left\{ \eta \in \{0,1\}^N : \sum_{x=1}^{N} \eta(x) = k \right\}, \qquad 1 \le k \le N/2.$$

$$(\mathcal{L}_{N,k} F)(\eta) = \sum_{x=1}^{N-1} c_x \big[F(\eta^{x,x+1}) - F(\eta) \big], \qquad r_x := c_x^{-1}.$$

$$\mu_{N,k}(\eta) = \binom{N}{k}^{-1}.$$

Say

At edge $\{x, x+1\}$ there is a clock of rate c_x. When it rings, we swap the two occupations. A particle next to a vacant site moves; two equal occupations do not change. The number of particles is conserved.

Say

Notice an important feature: the rate for a transition and its reverse is the same. Therefore the uniform measure on configurations with k particles is reversible. The environment changes the dynamics, but it does not change equilibrium. This is edge disorder with symmetric rates.

Say

I will assume $k \le N/2$ because exchanging the roles of particles and vacant sites gives the other case. The conductances may depend on N; I will suppress that superscript. Once the environment is fixed, this is an ordinary finite-state Markov chain.

Delivery cue. Spend about one minute on the generator. Point out the exchange $\eta^{x,x+1}$ rather than giving a separate formal definition. Do not introduce spectral gaps or a list of earlier results yet.

Transition. *Since the equilibrium measure is so simple, the real question is how long the dynamics takes to look uniform.*

BOARD 2 / 05–11 MINUTES

The theorem, with the assumptions explained

Say

Total variation measures how well an observation can distinguish the current configuration from equilibrium. Cutoff means that this distinguishability disappears abruptly when time is viewed on the scale of the mixing time.

WRITE ON THE BOARD

$$d_{N,k}(t) = \max_{\eta \in \Omega_{N,k}} \|P_t^{\eta} - \mu_{N,k}\|_{\mathrm{TV}}, \qquad t_{\mathrm{mix}}(\varepsilon) = \inf\{t : d_{N,k}(t) \le \varepsilon\}.$$

1. Averaging of resistance:

$$\delta_N := \frac{1}{N} \max_{1 \le m \le N} \left| \sum_{x=1}^{m-1} r_x - (m-1) \right| \longrightarrow 0,$$

2. Quantitative bounds on individual resistances:

$$\Upsilon_N \le r_x \le C_P \exp((\log N)^{\upsilon}), \qquad 0 < \upsilon < 1,$$

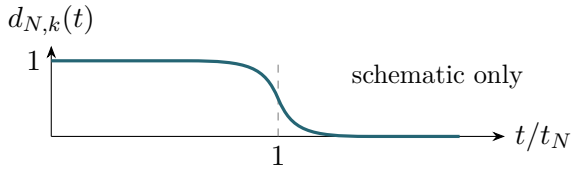
$$\Upsilon_N \to 0, \qquad \Upsilon_N \log N \to \infty.$$

3. Enough particles: $c_* N^{\gamma} \le k \le N/2$, $\gamma \in (0, 1]$, $c_* > 0$.

Theorem (Yang). For every fixed $\varepsilon \in (0, 1)$,

$$\boxed{t_{\mathrm{mix}}(\varepsilon) \sim t_N := \frac{N^2}{2\pi^2} \log k.}$$

Draw a schematic cutoff curve



Say

The first assumption says that total resistance up to a macroscopic location is approximately its length. Individual resistances may still vary substantially. The second group limits very large resistances and prevents very small resistances from approaching zero too fast. The last assumption asks that the particle number grow at least as a power of N.

Say

For a first example, take i.i.d. resistances in a fixed interval $[a, b] \subset (0, \infty)$, with mean one. Almost every environment satisfies the assumptions. The theorem also permits environments whose slowest rate tends to zero, and it applies to deterministic arrays satisfying these conditions.

Delivery cue. Explain each group as you write it; $C_P > 0$ is fixed. These are sufficient hypotheses, not a claimed sharp boundary of the cutoff regime. The graph is a schematic, not a limiting-profile assertion.

Transition. *The formula is familiar from the homogeneous model. Let me explain precisely what is added by this work.*

BOARD 3 / 11–15 MINUTES

What is the contribution?

WRITE ON THE BOARD

Known: homogeneous lower bound / cutoff (Wilson / Lacoin);
exclusion gap = random-walk gap (Caputo–Liggett–Richthammer).

This work: sharp cutoff with spatially varying conductances:

same leading constant + control of the disordered dynamics.

Key input: eigenvalue + eigenfunction shape + weighted derivative.

Say

For homogeneous conductances, Wilson identified the sharp lower-bound scale and Lacoin established cutoff. Also, the equality between the exclusion gap and the random-walk gap is already known for weighted graphs. The contribution here is a sharp total-variation mixing theorem for the inhomogeneous system: under the assumptions just stated, the transition remains at $N^2 \log k/(2\pi^2)$.

Why this requires work

Say

The stationary distribution stays uniform, but the dynamical calculations change. The ordinary cosine is no longer an exact eigenfunction, and the rates of the corner flips vary from site to site. Bounds using only the fastest and slowest rates lose information needed for the sharp constant. We need estimates that see how the environment averages, while still controlling individual updates.

Three contributions to emphasize

1. **The cutoff conclusion.** A class of inhomogeneous environments has the same leading mixing time as the homogeneous system, with convergence for every fixed accuracy ε.
2. **The spectral estimates supporting it.** The companion spectral paper supplies a deterministic Neumann analysis with explicit cosine approximation and uniform control of the weighted derivative. These estimates are precise enough for the many-particle proof.
3. **The proof in the disordered environment.** Fluctuation estimates and a coupling functional built from the environment's eigenfunction allow the lower and upper bounds to meet at the sharp time.

A useful sentence to say aloud.
The central issue is to carry information about one particle all the way to a sharp mixing theorem for the interacting system, without losing the leading constant.

Attribution for the speaker. The slides combine the cutoff paper and its companion spectral work. Spectral convergence has earlier foundations, including Faggionato's work. The eigenfunction method and height-coupling ideas also have predecessors. Emphasize the precise estimates and their successful use with inhomogeneous rates; do not claim invention of those general methods.

Transition. *I will now show why the proof needs three pieces of information about the slow mode, rather than only its decay rate.*

BOARD 4 / 15–20 MINUTES

The slow mode: rate, shape, and local flux

Say

When $k = 1$, we have a random walk with reflecting endpoints. Let λ_1 be its first nonzero eigenvalue, and normalize its corresponding eigenfunction by $g(1) = 1$. The principal eigenfunction here means the first nonconstant one.

WRITE ON THE BOARD

$$\mathcal{L}_{N,1} g = -\lambda_1 g, \qquad g(1) = 1, \qquad \sum_{x=1}^{N} g(x) = 0.$$

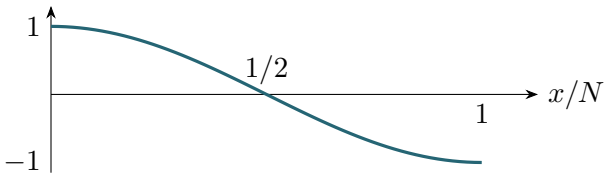
Spectral theorem. Under $\delta_N \to 0$ alone,

$$\lambda_1 \sim \frac{\pi^2}{N^2}, \qquad \max_{1 \le x \le N} |g(x) - h_N(x)| \longrightarrow 0, \qquad h_N(x) = \cos \frac{\pi(x - 1/2)}{N},$$

$$\max_{1 \le x < N} N \, |c_x[g(x+1) - g(x)] - [h_N(x+1) - h_N(x)]| \longrightarrow 0.$$

Also: g is strictly decreasing for every positive environment.

Draw the mode



Say

So the macroscopic shape is still a cosine. More is true: the conductance times the discrete derivative is also controlled uniformly. This is the flux. The unweighted derivative need not look locally homogeneous, because a large resistance can create a larger local drop.

Explain what each estimate buys

Gap λ_1	How fast does the remaining signal decay?
Shape of g	How large is the signal when particles start on the left?
Flux $c_x \Delta g$	How large are individual jumps, and which weights work?

Say

This combination is the useful spectral contribution for the cutoff problem. Closeness of two functions does not, on its own, control their discrete derivatives at the required scale. The weighted derivative must be controlled as well. All these estimates hold under resistance averaging alone, without independence or uniform bounds.

Delivery cue. If asked about many-particle spectral gaps, say that the exclusion gap equals the one-particle gap. The talk does not need a proof of that identity: the lower bound uses an explicit lifted eigenfunction, and the upper bound uses coupling.

Transition. *Why do resistances average, and why does the cosine appear? Both are visible in the paper's shooting recurrence.*

BOARD 5 / 20–27 MINUTES

The Riccati argument: why a cosine survives

Say

For a trial eigenvalue λ, start with $f(0) = f(1) = 1$ and solve the eigenvalue equation from left to right. An eigenvalue is a value for which the right endpoint also has zero flux.

WRITE ON THE BOARD

Flux: $J_x := c_x[f(x+1) - f(x)]$, $J_0 = 0$, $\lambda = \alpha/N^2$.

$$J_x - J_{x-1} = -\lambda f(x), \qquad J_x = -\lambda \sum_{y=1}^{x} f(y).$$

Flux divided by function: $B_x := -NJ_{x-1}/f(x-1)$, $B_1 = 0$.

$$\boxed{B_{x+1} = \frac{B_x}{1 - r_{x-1}B_x/N} + \frac{\alpha}{N}.}$$

Averaged equation, $s \approx x/N$**:**

$$B' = B^2 + \alpha, \qquad B(0) = 0, \qquad B(s) = \sqrt{\alpha}\tan(\sqrt{\alpha}\, s).$$

Across the pole: $A = 1/B$, $A' = -1 - \alpha A^2$.

Right boundary: $B_{N+1} = 0 \quad \rightsquigarrow \quad \sqrt{\alpha} = \pi$ for the first nonconstant mode.

Derive the recurrence in words

Say

First, $f(x)/f(x-1) = 1 - r_{x-1}B_x/N$. Next, the flux equation gives $-NJ_x = -NJ_{x-1} + \alpha f(x)/N$. Dividing by $f(x)$ gives exactly the boxed recurrence. We have converted a second-order equation into a first-order nonlinear recurrence.

Explain the two obstacles

Say

The averaging is not pointwise: an individual resistance need not be close to one. The proof uses the small error in cumulative resistance to control the accumulated error in the recurrence. Away from a pole, its limit is the Riccati equation on the board.

Say

A pole occurs when the eigenfunction crosses zero. The eigenfunction itself is regular; only this ratio blows up. We switch to the reciprocal variable, pass through its zero, and switch back. Tracking this passage selects the first mode. The tangent is interpreted branch by branch.

Recover the eigenfunction and its flux

Away from a zero, $g(x) = [1 - r_{x-1}B_x/N]g(x-1)$. The limiting equation $u' = -Bu$, with $u(0) = 1$, gives $u(s) = \cos(\pi s)$. The reciprocal variable handles the zero. Finally, $J_x = -\lambda_1 \sum_{y \le x} g(y)$ gives the uniform flux estimate.

Delivery cue. Spend two minutes on the recurrence, three on averaging and the pole, and two on the boundary and shape. Detailed error estimates are in Section 3 and Backup C. Use artificial $c_0 = c_N = 1$.

Transition. *We can now use this slow mode to detect memory in the particle system.*

BOARD 6 / 27–34 MINUTES

Lower bound: a signal larger than its fluctuations

WRITE ON THE BOARD

$$F(\eta) := \sum_{x=1}^{N} \eta(x) g(x), \qquad \mathcal{L}_{N,k} F = -\lambda_1 F, \qquad \mathbb{E}_\nu F(\eta_t) = e^{-\lambda_1 t} \nu(F).$$

Choose a left-biased initial law ν:

$$\nu(F) \geq ck, \quad \mathrm{Var}_\nu F(\eta_t) \leq Ck, \qquad \mu_{N,k}(F) = 0, \quad \mathrm{Var}_{\mu_{N,k}} F \asymp k.$$

$$\underbrace{ke^{-\lambda_1 t}}_{\text{remaining signal}} \asymp \underbrace{\sqrt{k}}_{\text{fluctuation size}} \iff \boxed{t = \frac{\log k}{2\lambda_1} \sim \frac{N^2 \log k}{2\pi^2}.}$$

First: an exact identity despite interaction

Say

The weight g is positive on the left and negative on the right. An exchange across an edge changes F by the occupation difference times the difference of the two weights. Summing over edges gives $\sum_x \eta(x) \mathcal{L}_{N,1} g(x) = -\lambda_1 F(\eta)$. So its expectation decays exactly at the one-particle eigenvalue.

Second: choose a start that allows fluctuation control

Say

For $k \geq N/64$, start with all particles on the left. For fewer particles, use the construction in the slides: sample $2k$ occupied sites uniformly and keep the leftmost k. The sample is spread out, but keeping its left half creates a signal of order k. This is the useful extra choice in a varying environment.

Say

To see why it helps, run the retained particles inside the stationary $2k$-particle system using the same edge exchanges. At every time the occupation probability of the smaller system is at most $2k/N$. A large resistance is therefore encountered with a controlled probability. Combined with the flux bound $|J_x| \leq C/N$ and total resistance of order N, this bounds the expected quadratic-variation rate by Ck/N^2. The martingale calculation gives variance of order k.

Third: turn signal separation into total variation

Let $m_t = e^{-\lambda_1 t} \nu(F)$. Chebyshev's inequality, applied to $\{F \geq m_t/2\}$, gives

$$d_{N,k}(t) \geq \|\nu P_t - \mu_{N,k}\|_{\mathrm{TV}} \geq 1 - \frac{Ck}{m_t^2}.$$

At $t = (1-\varepsilon)t_N$, $m_t/\sqrt{k} \geq ck^{\varepsilon/2+o(1)} \to \infty$, hence $d_{N,k}(t) \to 1$. A random initial law is legitimate because worst-case distance dominates the distance of any mixture of initial states.

Delivery cue. Pause at the boxed time: the factor $1/2$ comes from k versus $\sqrt{k}$. The complete variance and initial-law estimates are in Backup B, with no extra bounded-resistance assumption.

Transition. *This proves that we cannot mix earlier. To show that every initial state has mixed just after this time, we need a coupling argument.*

BOARD 7 / 34–40 MINUTES

Upper bound: weights that fit the environment

WRITE ON THE BOARD

$$H_\eta(x) := \sum_{y=1}^{x} \eta(y) - \frac{kx}{N}, \qquad H_\eta(0) = H_\eta(N) = 0.$$

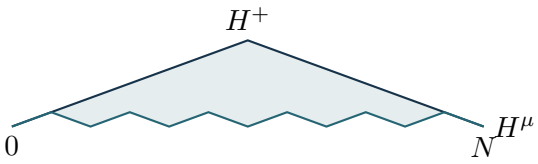
$$\eta \preceq \xi \iff H_\eta(x) \le H_\xi(x)\ \forall x; \qquad H_t^- \le H_t^\mu \le H_t^+.$$

Use the first eigenfunction G of a slightly longer segment:

$$w_x := G(x) - G(x+1) > 0, \qquad A_t := \sum_{x=1}^{N-1} w_x (H_t^+(x) - H_t^\mu(x)) \ge 0.$$

$$\operatorname{drift}(A_t) \le -\bar{\lambda}_1 A_t, \qquad \bar{\lambda}_1 \sim \frac{\pi^2}{(1+2\delta)^2 N^2}.$$

Draw the geometry



One possible pair for $N = 12$, $k = 6$. The shaded region is the discrepancy; A_t weights its vertical columns.

Say

The height at x measures excess particle mass to the left of x. Putting every particle at the left gives the maximal path; putting them at the right gives the minimal path. The process has a coupling that preserves this order. We compare an extremal path to a path already at equilibrium and ask when they coincide.

Say

A useful area must reflect the environment. We use the successive drops of its decreasing eigenfunction as weights. They are positive, and the eigenvalue equation gives a negative drift proportional to the area. The task is to control both the drift and the fluctuations with these varying weights.

Say

We choose the eigenfunction on a slightly enlarged interval, adding about δN sites at either end with unit conductances. The original interval then lies away from the reflecting endpoints, where the cosine slope vanishes. The flux estimate supplies useful positive weights throughout the original segment. We pay only a factor approaching one in the relaxation time as δ tends to zero.

Delivery cue. The extension follows an idea of Wilson. Stress the control of its weights and fluctuations in the disordered setting. The monotone coupling and boundary-term calculation are in Backup D.

Transition. *Negative drift gives the right decay scale. To get the sharp factor $1/2$, we must also use fluctuations to eliminate the remaining area.*

BOARD 8 / 40–43 MINUTES; QUESTIONS / 43–45

Finish the proof and return to the contribution

Say

There is a genuine extra step here. Exponential decay of the expected area alone is not sharp enough to force the area to be exactly zero. Near equilibrium, the randomness of the corner flips becomes essential.

State the remaining work in three sentences

The drift first brings the weighted discrepancy to a small fluctuation scale. Estimates on the height paths ensure enough available flips; quadratic-variation bounds and stopping times then force the remaining discrepancy to disappear. These are the technical parts of the upper-bound proof; they are not consequences of the spectral gap alone.

WRITE ON THE BOARD

$$T_+ := \inf\{t : H_t^+ = H_t^\mu\}, \qquad T_- := \inf\{t : H_t^- = H_t^\mu\}.$$

Technical coupling conclusion: for every fixed $\varepsilon > 0$,

$$\mathbb{P}(T_+ > (1+\varepsilon)t_N) + \mathbb{P}(T_- > (1+\varepsilon)t_N) \longrightarrow 0.$$

$$d_{N,k}(t) \le \mathbb{P}(T_+ > t) + \mathbb{P}(T_- > t).$$

$$\boxed{d_{N,k}((1-\varepsilon)t_N) \to 1, \qquad d_{N,k}((1+\varepsilon)t_N) \to 0} \quad (0 < \varepsilon < 1).$$

Say

For the coupling estimate we first fix a small enlargement, carry out the estimates, and then let the enlargement tend to zero. Once both extremal paths agree with the equilibrium path, every initial configuration has coupled as well. The coupling inequality now gives the upper bound. Together with the observable argument, this proves cutoff.

Closing words, approximately 45 seconds

The main contribution is a sharp cutoff theorem for this class of inhomogeneous conductances, with the same leading constant as in the homogeneous model. The proof uses the shape and weighted derivative of the actual slow mode to control fluctuations and construct a useful coupling distance. The scale comes from reducing a signal of size k to equilibrium fluctuations of size $\sqrt{k}$. The work is to prove that this intuition remains valid for the interacting system with spatially varying rates.

Leave these three items on the board

$$\lambda_1 \sim \pi^2/N^2, \qquad k\exp(-\lambda_1 t) \asymp \sqrt{k}, \qquad t_N = \frac{N^2}{2\pi^2}\log k.$$

Delivery cue. At minute 43, stop and invite questions. Leave the theorem and its contribution visible. Save higher eigenfunctions and other random environments for discussion.

BACKUP A / NOT PART OF THE 43-MINUTE PRESENTATION

Full hypotheses and normalization

The full cutoff statement in the supplied slides

Write $r_x^{(N)} = 1/c_x^{(N)}$. Assume

$$\frac{1}{N} \max_{1 \le m \le N} \left| \sum_{x=1}^{m-1} r_x^{(N)} - (m-1) \right| \longrightarrow 0. \tag{LLN}$$

In addition, for some $\upsilon \in (0,1)$ and $C_P > 0$,

$$\max_{1 \le x < N} r_x^{(N)} \le C_P \exp\left((\log N)^{\upsilon}\right).$$

There exists a positive sequence Υ_N such that

$$\min_{1 \le x < N} r_x^{(N)} \ge \Upsilon_N, \qquad \Upsilon_N \longrightarrow 0, \qquad \Upsilon_N \log N \longrightarrow \infty.$$

Finally, for some $\gamma \in (0,1]$ and $c_* > 0$,

$$c_* N^{\gamma} \le k_N \le N/2.$$

Then for every fixed $\varepsilon \in (0,1)$,

$$\frac{t_{\mathrm{mix}}^{N,k_N}(\varepsilon)}{N^2 \log k_N/(2\pi^2)} \longrightarrow 1.$$

The familiar special case $a \le r_x \le b$, with fixed $0 < a \le b < \infty$, satisfies the individual resistance bounds above. Eventually choose $\Upsilon_N = (\log N)^{-1/2}$ and adjust finitely many terms if needed.

Why an i.i.d. resistance sequence satisfies (LLN)

For a single infinite sequence with $\mathbb{E}r_x = 1$, let $S_j = \sum_{x=1}^{j}(r_x - 1)$. The strong law gives $S_j/j \to 0$. For every $\epsilon > 0$, choose M so that $|S_j| \le \epsilon j$ for $j \ge M$. Then

$$\frac{1}{N} \max_{0 \le j < N} |S_j| \le \frac{1}{N} \max_{j < M} |S_j| + \epsilon \longrightarrow \epsilon.$$

Let $\epsilon \downarrow 0$. This explains the almost-sure, or quenched, interpretation. The spectral theorem also accommodates stationary ergodic resistances with finite mean, through the same partial-sum argument.

If the mean resistance is $m \neq 1$

Normalize $\widehat{r}_x = r_x/m$, so $\widehat{c}_x = mc_x$ and $\widehat{\mathcal{L}} = m\mathcal{L}$. Hence, under the corresponding assumptions,

$$\lambda_1 \sim \frac{\pi^2}{mN^2}, \qquad t_{\mathrm{mix}}(\varepsilon) \sim \frac{mN^2}{2\pi^2} \log k.$$

Thus the effective conductance is $1/m$, the reciprocal of the mean resistance. The mean of c_x is generally not the coefficient governing the time scale.

BACKUP B1 / LOWER BOUND UNDER THE STATED ASSUMPTIONS

Initial bias and variance: the choice of starting law

Write $\lambda = \lambda_1$, $M = \|g\|_\infty = O(1)$ and $J_x = c_x[g(x+1) - g(x)]$. The spectral theorem and resistance averaging give

$$\lambda \asymp N^{-2}, \qquad \max_x |J_x| \le C/N, \qquad \sum_{x=1}^{N-1} r_x = O(N).$$

All constants below are uniform in N, k, t for large N. The particle-number hypothesis ensures $k \to \infty$.

Equilibrium fluctuations

Sampling k sites without replacement and using $\sum_x g(x) = 0$ gives

$$\mathrm{Var}_{\mu_{N,k}} F = \frac{k(N-k)}{N(N-1)} \sum_{x=1}^{N} g(x)^2 \asymp k,$$

since $N^{-1} \sum_x g(x)^2 \to 1/2$.

Dense regime: $k \ge N/64$

Take $\nu = \delta_{\eta^+}$, with $\eta^+(x) = \mathbf{1}_{\{x \le k\}}$. Then

$$F(\eta^+) = \sum_{x=1}^{k} g(x) = \frac{\sin(\pi k/N)}{2\sin(\pi/(2N))} + o(k) \ge ck, \qquad \mathrm{Var}_\nu F = 0.$$

Sparse regime: $k < N/64$

Sample $\zeta \sim \mu_{N,2k}$ and retain its leftmost k particles to obtain η with law ν. Put $\ell = \lfloor N/2 \rfloor$ and

$$X = \sum_{x \le \ell} \zeta(x), \qquad Z = \sum_{x \le \ell} \zeta(x) g(x).$$

The first ℓ sites contain X sampled particles; retaining the first k adds or removes exactly $|X - k|$ terms relative to Z. Hence

$$|F(\eta) - Z| \le M|X - k|.$$

The hypergeometric variance formula gives $\mathbb{E}(X-k)^2 \le Ck$, and the same finite-population calculation gives $\mathrm{Var}\, Z \le Ck$. Also,

$$\mathbb{E}Z = \frac{2k}{N} \sum_{x \le \ell} g(x) = \left(\frac{2}{\pi} + o(1)\right) k.$$

Consequently,

$$\nu(F) \ge \left(\frac{2}{\pi} + o(1)\right) k - C\sqrt{k} \ge ck,$$

$$\mathrm{Var}_\nu F \le 2\,\mathrm{Var}\, Z + 2\mathbb{E}(F - Z)^2 \le Ck.$$

This calculation explains why the random start is both left-biased and sufficiently concentrated. It uses only uniform eigenfunction approximation and elementary sampling identities.

BACKUP B2 / LOWER BOUND CONTINUED

Martingale fluctuations and the separating event

The quadratic-variation rate

An edge exchange changes F by $(\eta(x)-\eta(x+1))(g(x+1)-g(x))$. Thus

$$\Gamma F(\eta) := \mathcal{L}(F^2)(\eta) - 2F(\eta)\mathcal{L}F(\eta) = \sum_{x=1}^{N-1} r_x J_x^2 (\eta(x+1)-\eta(x))^2.$$

For $k \geq N/64$, this is at most $CN^{-2}\sum_x r_x \leq C/N \leq C'k/N^2$ for every configuration.

For $k < N/64$, couple η_0 from Backup B1 with its parent configuration $\zeta_0 \sim \mu_{N,2k}$. Use common edge swaps: coordinatewise inclusion $\eta_t(x) \leq \zeta_t(x)$ is preserved and ζ_t remains stationary. Therefore

$$\mathbb{E}_\nu \eta_t(x) \leq \frac{2k}{N}, \qquad \mathbb{E}_\nu(\eta_t(x+1)-\eta_t(x))^2 \leq \frac{4k}{N},$$

and

$$\mathbb{E}_\nu \Gamma F(\eta_t) \leq \frac{C}{N^2}\frac{4k}{N}\sum_x r_x \leq \frac{C'k}{N^2}.$$

This coupling uses coordinatewise inclusion for different particle numbers; the height-order coupling in the upper bound is a separate construction.

The time-dependent martingale

For fixed t, $e^{-\lambda(t-s)}F(\eta_s)$, $0 \leq s \leq t$, is a martingale. Orthogonality of its increments to the initial value gives

$$\begin{aligned}\operatorname{Var}_\nu F(\eta_t) &= e^{-2\lambda t}\operatorname{Var}_\nu F + \int_0^t e^{-2\lambda(t-s)}\mathbb{E}_\nu \Gamma F(\eta_s)\,ds \\ &\leq Ck + \frac{Ck}{2\lambda N^2} \leq C'k.\end{aligned}$$

Separate the law from equilibrium

Let $m_t = e^{-\lambda t}\nu(F)$ and $B_t = \{F \geq m_t/2\}$. By Chebyshev,

$$d_{N,k}(t) \geq (\nu P_t)(B_t) - \mu_{N,k}(B_t) \geq 1 - \frac{Ck}{m_t^2}.$$

At $t = (\log k)/(2\lambda) - a/\lambda \geq 0$, this gives $d_{N,k}(t) \geq 1 - Ce^{-2a}$. For fixed $\varepsilon > 0$, it also gives $d_{N,k}((1-\varepsilon)t_N) \to 1$.

Precision of centering. The first estimate is centered at the exact spectral time $(\log k)/(2\lambda_1)$. It does not establish an $O(N^2)$ cutoff window about $N^2\log k/(2\pi^2)$: a relative $o(1)$ error in the gap can shift that center by more than N^2.

BACKUP C / SPECTRAL PROOF DETAILS

The Riccati variable and the flux comparison

Exact shooting recurrence in the manuscript's indexing

Set $f^\lambda(0) = f^\lambda(1) = 1$ and impose the eigenvalue recurrence from the left. Use an artificial $c_N = 1$ for the last step. Write

$$(c\nabla f)(x) = c_{x-1}[f(x) - f(x-1)], \qquad b(\lambda, x) = -\frac{(c\nabla f^\lambda)(x)}{f^\lambda(x-1)}.$$

Where the ratio is finite,

$$b(\lambda, x+1) = \frac{b(\lambda, x)}{1 - r_{x-1} b(\lambda, x)} + \lambda.$$

At a zero of the denominator, interpret the ratio on the projective line. With $B_x = Nb(\lambda, x)$ and $\lambda = \alpha/N^2$,

$$B_{x+1} = \frac{B_x}{1 - r_{x-1} B_x/N} + \frac{\alpha}{N}, \qquad B_1 = 0.$$

The right reflecting condition is $B_{N+1} = 0$. The averaged equation is

$$B'(s) = B(s)^2 + \alpha, \qquad B(s) = \sqrt{\alpha}\tan(\sqrt{\alpha}\, s).$$

This tangent formula is used *between its poles*; it is not a finite classical solution throughout $[0, 1]$ when a pole lies inside. Near the zero of the eigenfunction, change to $A = 1/B$, whose limiting equation is

$$A'(s) = -1 - \alpha A(s)^2.$$

One then returns to B after the pole. Tracking the phase selects the first nonconstant mode rather than an arbitrary higher eigenvalue. This is the role of the first, middle, and last segments in the manuscript.

Why cumulative resistance suffices for the averaging step

Let $S_x = \sum_{y=1}^{x}(r_y - 1)$, so $\max_x |S_x| = o(N)$. For controlled weights a_x, summation by parts gives

$$\sum_{x=1}^{m}(r_x - 1)a_x = S_m a_m + \sum_{x=1}^{m-1} S_x(a_x - a_{x+1}).$$

After division by N, this is $o(1)$ when the weights have uniformly bounded size and total variation. The proof obtains the required control in the appropriate coordinate charts.

Flux convergence once the eigenvalue and shape are known

Let $\lambda_0 = 2(1 - \cos(\pi/N))$. For $1 \le x < N$,

$$J_x = -\lambda_1 \sum_{y=1}^{x} g(y), \qquad h_N(x+1) - h_N(x) = -\lambda_0 \sum_{y=1}^{x} h_N(y).$$

Consequently,

$$\max_{x<N} N|J_x - [h_N(x+1) - h_N(x)]| \le N^2\lambda_1 \|g - h_N\|_\infty + N^2|\lambda_1 - \lambda_0| \longrightarrow 0.$$

BACKUP D / UPPER-BOUND DETAILS

Why enlarge the interval? Why is drift insufficient?

The exact boundary terms

Let $m = \lfloor \delta N \rfloor$ and extend the segment to $\{-m, \dots, N+m\}$, of size $\bar{N} = N + 2m + 1$, keeping the original conductances and setting the added ones to one. Let G be its decreasing first eigenfunction and $\bar{\lambda}$ its eigenvalue. Define $w_x = G(x) - G(x+1)$ and

$$\Phi(\eta) = \sum_{x=1}^{N-1} w_x H_\eta(x).$$

The original height dynamics satisfies

$$\mathcal{L}H_\eta(x) = c_x[H_\eta(x+1) - 2H_\eta(x) + H_\eta(x-1)].$$

The extended eigenvalue equation implies

$$c_{x+1}w_{x+1} - 2c_x w_x + c_{x-1}w_{x-1} = -\bar{\lambda} w_x.$$

Summation by parts, using $H_\eta(0) = H_\eta(N) = 0$, yields

$$\mathcal{L}\Phi(\eta) = -\bar{\lambda}\Phi(\eta) - c_0 w_0 H_\eta(1) - c_N w_N H_\eta(N-1).$$

For an ordered pair $H^+ \geq H^\mu$, put $D_x = H^+(x) - H^\mu(x) \geq 0$. Its weighted discrepancy has drift

$$-\bar{\lambda}A - c_0 w_0 D_1 - c_N w_N D_{N-1} \leq -\bar{\lambda}A.$$

The lower extremal pair is handled identically, with the height difference reversed.

Why the extension is useful

The weighted derivative theorem on the enlarged interval gives

$$c_x w_x \asymp_\delta N^{-1} \qquad (1 \leq x < N).$$

The original edges are a fixed positive macroscopic distance from the enlarged reflecting endpoints, so the limiting sine factor stays positive there. Under $a \leq r_x \leq b$, this also gives $w_x \asymp_\delta N^{-1}$. The enlargement changes the gap only to $\bar{\lambda} \sim \pi^2/((1+2\delta)^2 N^2)$.

Why expectation decay alone loses the sharp time

We have $A_0 = O(k)$ and $\mathbb{E}A_t \leq Ck\exp(-\bar{\lambda}t)$. However, the smallest positive weighted discrepancy is of order $1/N$ in the bounded-resistance setting. Markov's inequality based only on the expectation would require decay to that microscopic scale. It does not give the desired $\frac{1}{2}\log k$ factor.

What is intentionally omitted from the talk

The full argument uses stopping levels of the area, quadratic-variation bounds, and controls on available corner flips and height fluctuations. Censoring supplies further height estimates. These turn small discrepancy into coalescence. See Section 5 and Appendices B–C of [3]; Board 8 states their coupling consequence.

Coupling construction. Use up and down clocks at each allowed site–height pair (x, z), each of rate c_x. A path responds only at its current height, making an allowed flip; paths at the same level share clocks. This preserves order. Common undirected edge swaps alone do not preserve height order.

BACKUP E / QUESTIONS

Answers to likely seminar questions

Why not average the conductances? The exact identity is $g(x+1) - g(x) = r_x J_x$. Resistance multiplies a flux that varies on the macroscopic scale. For mean resistance m, the time scale is multiplied by m.

Is the environment required to be random? No. Both stated conclusions are deterministic once their assumptions hold. Randomness is one way to obtain the resistance averaging condition; for an infinite i.i.d. environment it holds almost surely when the mean is finite.

Does the resistance law of large numbers alone prove cutoff? It suffices for the spectral theorem. The cutoff theorem also assumes the resistance bounds and polynomial particle-number condition on Board 2. The full lower-bound calculation in Backup B uses resistance averaging and the spectral estimates; the additional bounds enter the upper-bound analysis.

Why does knowing the spectral gap not settle the mixing time? The gap gives a decay rate, but sharp mixing also depends on the size of an initial signal, equilibrium fluctuations, and control of all other discrepancies. The lower-bound observable and the upper-bound coupling play different roles.

What if $k > N/2$? What if k is fixed? Use particle–vacant-site symmetry and replace k by $\min(k, N-k)$. The stated cutoff theorem does not cover a fixed number of particles; do not extend the formula to $k = 1$, where $\log k = 0$.

Do we know the window or the cutoff profile? The result presented gives the leading mixing time and cutoff. It does not by itself identify an $O(N^2)$ window around the deterministic leading term or a limiting profile. An $o(1)$ relative error in the gap can produce a centering error larger than N^2 after multiplication by $\log k$.

What is new in the spectral input for this application? Emphasize the complete set of estimates: cosine shape, uniform weighted derivative, and gap asymptotic, obtained through a deterministic one-dimensional proof. Their role is to make the interacting-particle argument work. Faggionato's earlier spectral convergence theory should be acknowledged; a gap asymptotic alone is not the novelty claim.

What should I say if someone asks: "Is this just the homogeneous proof?" The architecture builds on existing eigenfunction and coupling methods. The work is to replace explicit homogeneous formulas by estimates for the actual environment, and to retain the sharp constant through the fluctuation and coalescence bounds. The resulting theorem covers the full resistance assumptions, beyond a uniformly elliptic special case.

Can I call this a universality result? You can describe stability of the leading cutoff time across the class in the theorem. Keep the resistance and particle-number restrictions attached to that statement; the result does not cover arbitrary disorder.

BACKUP F / CONTRIBUTION AND ATTRIBUTION

An example showing the scope of the theorem

For sufficiently large N, set $r_x = 1$ except at two distinct edges:

$$r_1 = (\log N)^{-1/2}, \qquad r_{\lfloor N/2 \rfloor} = \log N.$$

Then $\delta_N = O(\log N/N) \to 0$. Choose $\Upsilon_N = (\log N)^{-1/2}$ and $\upsilon = 1/2$. The upper resistance bound holds eventually since $\log N \le \exp(\sqrt{\log N})$ for all sufficiently large N.

Thus one conductance diverges and another tends to zero, yet the cutoff theorem applies whenever the particle-number condition holds. This example follows directly from the hypotheses and illustrates the scope beyond uniformly bounded positive rates.

Keep the two papers' contributions distinct

Cutoff paper	Sharp many-particle mixing time and cutoff under inhomogeneous conductances.
Spectral paper	Deterministic Neumann eigenfunction and flux estimates, with a direct shooting/Riccati proof, supplying the spectral input.
Earlier foundations	Homogeneous cutoff; the spectral-gap identity; spectral convergence; eigenfunction and coupling techniques.

Sources and attribution

[1] **Main source:** S. Yang, *Cutoff of the simple exclusion process with inhomogeneous conductances.* Supplied `cutoff-SEP-RC(1).pdf`, 29 PDF pages (22 numbered slides). Slides 1–10 supply the model and theorems; 12–15 the spectral proof; 16–18 the lower bound; 19–22 the upper bound.

[2] **Supporting paper:** S. Yang, *The spectral gap and principal eigenfunction of the random conductance model in a line segment.* Supplied `AIHP-RCM.pdf`, 23 pages. Proposition 1.1: monotonicity; Theorem 1.3: gap, shape and weighted derivative; Remark 1.4: fixed higher modes. Section 1.4 explains the recurrence; Sections 3–4 contain its rigorous analysis. Source for Boards 4–5 and Backup C.

[3] S. Yang, *Cutoff of the simple exclusion process with inhomogeneous conductances*, arXiv:2409.16337. Section 5 and Appendices B–C of v1 support the description of the technical upper-bound estimates. The supplied slides govern the theorem formulation in these notes.

[4] H. Lacoin, *Mixing time and cutoff for the adjacent transposition shuffle and the simple exclusion*, Ann. Probab. **44** (2016), 1426–1487. D. B. Wilson, *Mixing times of lozenge tiling and card shuffling Markov chains*, Ann. Appl. Probab. **14** (2004), 274–325. These are predecessors of the homogeneous cutoff and proof methods.

[5] P. Caputo, T. M. Liggett and T. Richthammer, *Proof of Aldous' spectral gap conjecture*, arXiv:0906.1238. Background for the already-known gap identity.

[6] A. Faggionato, *Spectral analysis of 1D nearest-neighbor random walks and applications to subdiffusive trap and barrier models*, Electron. J. Probab. **17** (2012), no. 15; arXiv:0905.2900. Earlier spectral convergence framework.

Preparation note. The talk follows the first file's progression from cutoff to spectral input, lower bound, and upper bound. The initial-law and martingale calculations in Backup B are written out for seminar use. The Riccati limit is presented as a proof sketch; the technical coalescence step is explicitly stated as an input.